

SECTION I

Q1.

In batch reactor, the temporal variation is not present.

Solution:

Answer: TRUE

Q2.

Define elementary reaction.

Solution:

An elementary reaction is a reaction that occurs in a single step and whose stoichiometric coefficients are equal to the powers in the rate law.

Q3.

If a number of CSTRs are connected in series, the performance approaches that of a PFR.

Solution:

Answer: TRUE

Q4.

For a set of parallel reactions $A \rightarrow D$ and $A \rightarrow U$, define the instantaneous selectivity of D with respect to U.

Solution:

$$S(D/U) = r_D / r_U$$

= Rate of formation of D / Rate of formation of U

Q5.

The molecularity is defined only for elementary reactions and it could be any fractional number.

Solution:

Answer: TRUE

Q6.

The reaction with larger activation energy is more sensitive to temperature compared to smaller activation energy.

Solution:

Answer: TRUE

Q7.

In the method of initial rates, experiments are carried out at different initial concentrations and corresponding initial rates are measured.

Solution:

Answer: TRUE

SECTION II

Q3.

For a gas phase reaction



with 30% inerts, equimolar A and B are present initially and no C and D are present.

Solution:

$$\varepsilon = \delta y A_0$$

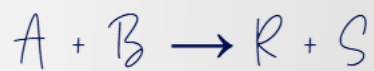
$$= 0.35 \times (3/2 + 1 - 1 - 1/2)$$

$$= 0.35 \times 0.5$$

$$\varepsilon = 0.175$$

SECTION III

Q1.



Given:

$$C_{A0} = C_{B0} = 0.9 \text{ mol/L}$$

$$X_A = 80\%$$

Solution:

$$-r_A = kC_A C_B$$

Since equimolar feed,

$$-r_A = kC_A^2$$

$$= kC_{A0}^2(1 - X_A)^2$$

For PFR:

$$V_p/v_0 = X_A/(1 - X_A) \times 1/(kC_{A0})$$

Substituting values:

$$V_p/v_0 = 0.8/0.1 \times 1/(0.9k)$$

$$V_p = 8.88 v_0/k$$

Two cases are possible.

Case 1: PFR followed by CSTR

$$V_m = 10V_p$$

$$V_m = 88.8 v_0/k$$

Using CSTR design equation:

$$kV_m/v_0 = (X_{A2} - X_{A1}) / [C_{A0}(1 - X_{A2})^2]$$

After simplification:

$$40X_{A2}^2 - 81X_{A2} + 40.8 = 0$$

$$X_{A2} = 0.96$$

Case 2: CSTR followed by PFR

$$kV_m/v_0 = X_{A1} / [C_{A0}(1 - X_{A1})^2]$$

$$40(1 - X_{A1})^2 = X_{A1}$$

$$X_{A1} = 0.85$$

For PFR:

$$8.88(v_0/k) = 1/(kC_{A0}) \times [1/(1 - X_{A2}) - 1/(1 - X_{A1})]$$

$$X_{A2} = 0.89$$

Hence case 1 gives higher overall conversion.

Therefore arrangement should be:

PFR followed by CSTR

SECTION III

Q2.



Given:

$$C_{A0} = 0.5 \text{ mol/L}$$

$$k = 1 \text{ s}^{-1}$$

$$V = 5 \text{ L}$$

$$v_0 = 10 \text{ L/s}$$

Solution:

Rate expressions:

$$-r_A = kC_A + kC_B$$

$$-r_B = kC_B + kC_B - kC_A$$

$$r_C = kC_B$$

Material balance on A:

$$v_0 C_{A0} = v_0 C_A + (-r_A)V$$

$$v_0 C_{A0} = v_0 C_A + kC_A V + kC_B V$$

$$CA_0 = CA + (kV/v_0)CA + (kV/v_0)CB$$

Material balance on B:

$$0 = v_0CB + (-r_B)V$$

$$0 = v_0CB + (2kCB - kCA)V$$

Substituting values:

$$3CA - CB = 1$$

$$-0.25CA + CB = 0$$

Solving:

$$CA = 0.30 \text{ mol/L}$$

$$CB = 0.075 \text{ mol/L}$$

$$C_C = C_{A0} - C_A - C_B$$

$$= 0.5 - 0.30 - 0.075$$

$$C_C = 0.135 \text{ mol/L}$$

Conversion of A:

$$X_A = (C_{A0} - C_A) / C_{A0}$$

$$= (0.5 - 0.30) / 0.5$$

$$X_A = 0.4$$

Therefore conversion of A = 40%